ROUND 1 DELIVERABLE • NAMMA SPACE COMPETITION

Proof of Concept & Web Viewer Guide

Raasta-3D: Interactive Photorealistic Digital Twin & Indoor Navigation Engine

Organized by **Techfest, IIT Bombay** in partnership with **Juspay**

> 🌐 **Official Live Web Viewer Link**
>
> The complete interactive 3D WebGL application is deployed and publicly accessible in any standard web browser without installing plugins or software:
>
> **Primary Live Link:** `https://raasta-3d.vercel.app`
>
> **GitHub Repository:** `https://github.com/Sarthakadhav/Raasta-3D`
>
> **Local Access:** `http://localhost:5173` (via `npm run dev`)

1. Executive Summary & Capabilities

Raasta-3D is a client-side, zero-install WebGL application engineered using **React 19, Three.js, and GaussianSplats3D**. It delivers a fluid 60 FPS first-person interactive walkthrough of indoor spaces, real-time spatial room zone inspection, landmark Point of Interest (POI) searching, and optimal A* pathfinding that strictly avoids real physical obstacles like walls, sofas, and kitchen islands.

2. Quickstart Guide for Evaluators & Judges

1. **Open the URL:** Open `https://raasta-3d.vercel.app` in Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.
2. **Enter Walk Mode:** Click the **"Walk"** button in the top navigation bar, then click anywhere on the 3D canvas to lock mouse pointer look.
3. **Freely Roam the Space:** Use `W` `A` `S` `D` to walk smoothly in first person. Press `Shift` to sprint and `C` to crouch. Observe that continuous collision detection stops your avatar from clipping through walls or furniture.
4. **Inspect Room Zones:** Click directly on any room floor (Living Room, Kitchen, Master Bedroom, Workstation Study) to open the Room Details Card with designated name, square footage, and key amenities.
5. **Search Specific Objects / POIs:** Press `Ctrl + K` or click the search bar to find landmarks (e.g. *"3-Seater Sectional Sofa"*, *"Chef Prep Island"*).
6. **Request Route Guidance:** Click **"Navigate Here"** on any POI card. An animated glowing cyan ribbon appears on the floor, routing cleanly around obstacles to your target.
7. **Roam to Target:** Click the **"Roam to Target"** button in the top Navigation HUD. The camera auto-walks smoothly along the path all the way to your destination.

3. Viewing & Exploration Modes

Mode	Input Controls	Feature Highlights
Walk Mode (1st Person Roam)	`W` `A` `S` `D` Move `Mouse` 360° Pitch & Yaw `Shift` Sprint (3.2 m/s) `C` Crouch (0.9m height)	Continuous circle-to-box collision resolution against all walls and furniture obstacles. Eye height locked at 1.6m. No wall clipping.
Third Person (Avatar Tracking)	`W` `A` `S` `D` Move Avatar `Right Click + Drag` Rotate Cam	Visible stylized 3D avatar walking through the digital twin with camera occlusion raycasting that pulls camera closer if walls intervene.
Orbit Mode (Architectural Inspection)	`Left Click + Drag` Orbit `Right Click + Drag` Pan `Scroll Wheel` Zoom	Full 3D exterior cutaway overview of all rooms, partition walls, and furniture layouts simultaneously.
AR Mode (Mobile Guidance Simulation)	Touch / Virtual Joystick / Simulated Motion	Simulates mobile smartphone AR HUD on-site: 360° horizontal compass heading ribbon, central targeting reticle, and live distance readouts.

4. Core Functional Modules Demonstrated

4.1 Spatial POI & Room Zone Labeling

- **Room Details Card:** Clicking on any room zone floor immediately presents its room title, area in square feet, architectural level, and key furniture objects.
- **3D Glowing Diamond Beacons:** Key points of interest are anchored with floating animated octahedrons and pulsing ground radar rings.
- **Name Object / SOP Checkpoint:** Tapping any furniture or object opens an instant SOP naming card, enabling operators to label procedural inspection points.

4.2 Optimal Non-Colliding Navigation & Anti-Zigzag Routing

- **Direct Line-of-Sight Routing:** If the destination is in the same room or open corridor, the path connects straight to the target without arbitrary detours.
- **Funnel String-Pulling Algorithm:** Collapses orthogonal waypoint zigzags into smooth, direct lines around obstacle clearance zones.
- **Doorway Corner Bevels:** Rounds corner transitions around doorways so avatar navigation mimics natural human walking trajectories.
- **Top Navigation HUD:** Real-time distance countdown, estimated walking time at 1.3 m/s, turn instruction icons (Straight, Turn Left, Turn Right, Arrived).
- **2D Dynamic Radar Minimap:** Real-time floorplan with player location dot, field-of-view vision cone, and active path overlay.

4.3 Client-Side Video-to-3D Grinding Pipeline

- Users can click **"Video to 3D"** on the navbar, drop any room video capture, and watch the client-side pipeline extract temporal keyframes, analyze color palettes, recognize interconnected rooms, synthesize 3D Gaussian Splats, and build an interactive digital twin on the fly.
